

Modeling Report: Predicting Freight Rates

The Big Picture

We're building a model to predict freight rates for the end of the year—specifically from November 1 to December 31, 2025. To get there, we used historical data from the first ten months of 2025. One thing we did differently was how we checked our work. Instead of just picking random spots to test, we used a time-based strategy. This means we trained the model on the past and tested it on the "future" to make sure it can actually handle real-world forecasting without cheating by looking at data it shouldn't have yet.

What I sFound in the Data

- **How much data:** We looked at 48,000 different entries in our training set.
- **The main driver:** Distance is the biggest factor by far, with a 0.91 correlation to the rate.
- **Seasonal swings:** Rates definitely change with the seasons, climbing from about \$2.10 in January to \$2.33 in June.
- **Equipment matters:** Reefer loads are the priciest (\$2.38), followed by Flatbeds (\$2.29) and Dry Vans (\$2.12).
- **Messy details:** We ran into some issues with missing or negative *weight* values, missing *market_index* stats, and some wonky GPS coordinates for cities like Shreveport and New Orleans.
- **The challenge:** Our test set included 8 cities and 736 lanes that weren't in the training data at all, so our model had to be smart enough to generalize.

Cleaning Up the Mess

- **Weight:** We fixed 292 negative numbers and filled in 300 blanks using the middle-of-the-road weight of 31,496.
- **Market Index:** We plugged 374 missing holes with the median value of 1.0558.
- **Distance:** Since the coordinates were a bit flaky, we stuck with the provided *distance* column to keep things accurate.

How I Tested It

Mixing the data up randomly doesn't work here because it lets the model "see" into the future, which isn't how things work in the real world. Instead, we split the data chronologically: we

trained on the early part of the year and tested on the most recent months. This is a much better way to see how the model will actually perform when we ask it to forecast next month's rates.

How Well Did It Do?

Model	MAE (\$)	RMSE (\$)	MAPE
Naive rate-per-mile	290.60	709.45	12.13%
XGBoost (log target)	145.79	645.82	6.98%